

Recruitment task

Software Developer

The goal is to create an application for Microsoft Windows, that will be able to display an overlay on the window of a specific game (that is - a few interface controls). On top of that it should allow to interact with these controls, altering the way the game handles input at that time.

Selected game

For the purpose of the task - a GOG.com game will be chosen. The game key will be provided when the test starts. The application which will be created as a result of the task, will be tested only with selected game.

Control types

The application overlay should use the following interface controls:

- *sprite* - static control (no interaction), stateless. Its only task is to display a static image, loaded from a file.
- *label* - static control (no interaction), stateless. Its only task is to display text.
- *button* - interactive control, displaying text and allowing to trigger an action as an effect of being pushed by a user (button press); Button control can be in one of the following states:
 - *neutral*
 - *hovered* (mouse over the control)
 - *active* (LMB pressed within control bounds)

Button states can be visualized as different colors - it is not needed to differentiate the states with images.

Input

The test application, apart from displaying the overlay, should also be able to react to user input actions (mouse and keyboard). In particular - pressing a specific key combination should be handled by the test application, and not by the game. The same scenario applies to the mouse.

Requirements: test scenario

If the test application is started at the moment the game is not running - the application should terminate.

The application should start and enter the state **overlay disabled**, when executed at the time the game is running. When in this state, in the bottom-right corner of the game view - control of type *label* is being displayed with the text: "*Ctrl+space to enable overlay*".



Overlay enabled

Pressing CTRL+SPACE makes the application change its state to **overlay enabled**.

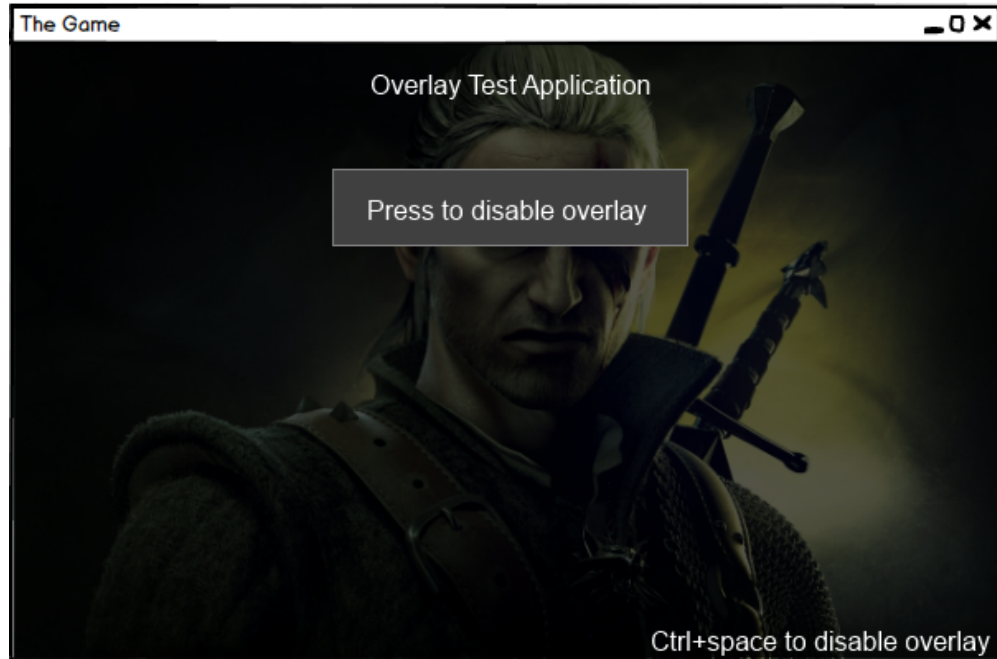
When in this state, the application displays the following elements:

- sprite control, displaying a half-transparent image, which darkens the entire game view,
- label control, displaying text "Ctrl+space to disable overlay" in the bottom-right corner of the game window,
- label control in the upper-center part of the game view, displaying text: "Overlay Test Application",
- button control just below. Text on the button: "Press to disable overlay". Pressing the button should switch the state to **overlay disabled**.

Pressing CTRL+SPACE, when in **overlay enabled** state should also switch the state to **overlay disabled**.

When in **overlay enabled** state, the application captures all the user input from the mouse and keyboard, so that the input events are not passed to the game.

The test application should never pause or freeze the game process, or alter it in any other way, regardless of the current state.



Overlay enabled

Additional requirements

The test application, when in **overlay enabled** state should use a system arrow cursor. When in **overlay disabled** state - it should also properly handle the following situations:

- resolution change
- fullscreen / window mode
- game close (the application should terminate)

Technology

The test application should be a Windows console application, written in C++. Usage of commonly available, free to use libraries is allowed.

Duration

The task should be done in 3 days.

Evaluation criteria

The task will be evaluated using the following criteria (the first being the most important):

1. Fulfilling the requirements described in the test scenario,
2. Code quality - no warnings, readability (self-commenting, comments), naming convention, code design, performance,
3. Fulfilling additional requirements,
4. Meeting the deadline,
5. Invention - all other things that improve the application in any way, not described by the requirements or additional requirements.